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a

Business Proposal

on

AI-Driven Predictive & Blockchain-Secured Food Waste Management Platform

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in

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in

INFORMATION SYSTEMS AND PROCESSES

Submitted by

MERUVA VIJAYA RAMASESHU (W20115923)

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AI-Driven Predictive & Blockchain-Secured Food Waste Management Platform

Date: October 2025

Executive Summary:

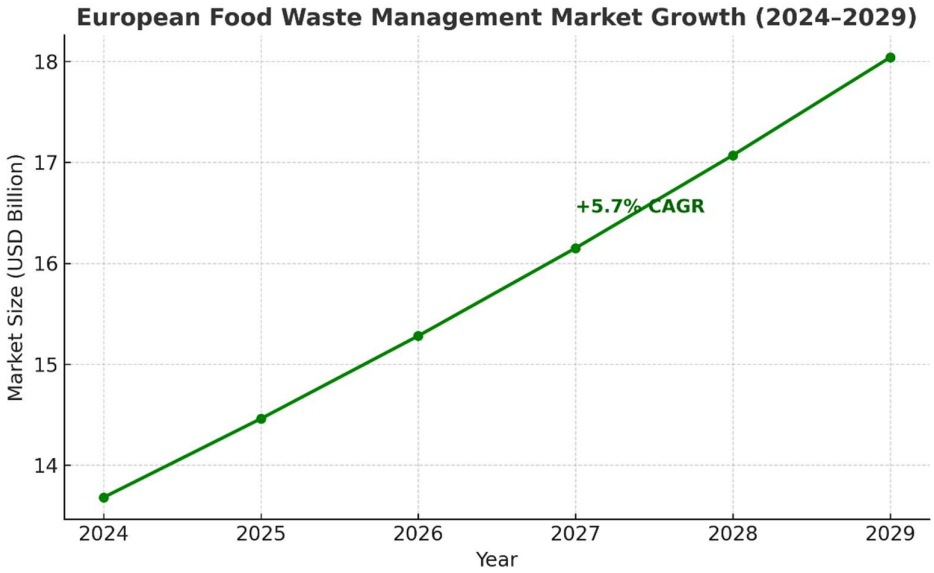
The concept emerged as a result of growing awareness of food waste as a serious environmental issue and economic problem in Europe. In the European Commission alone, almost 58 million tonnes of food is wasted annually and cost it more than EUR132 billion [1]. The management structures that are in existence today lack transparency, accountability and predictive efficiency. Our solution integrates blockchain and artificial intelligence to create a sustainable, data-driven ecosystem to minimize waste, promote traceability, and promote the goals of sustainability of the EU (based on SDG 12: Responsible Consumption and Production).

Introduction & Background

One of the most significant issues that can be considered both a sustainability and economic challenge in Europe is the food waste. In European market, the wasted food amounts to over 59 million tonnes annually that is equivalent to 132 kg per individual [2]. The solution proposed delivers smarter redistribution, transparency, and scalability throughout the supply chain with the help of AI, blockchain, and cloud computing.

Market Overview:

The European food waste management sector is projected to grow gradually with an average of 5.7 percent/year until 2029, currently standing at USD 13.68 billion in 2024[3]. The reason behind this increase is the growing environmental trends, the shift towards green digital technologies, and stricter EU regulations that lead to increased adoption of AI-based fixes aimed at reducing wastes and offering new possibilities in the field of the circle economy [4].



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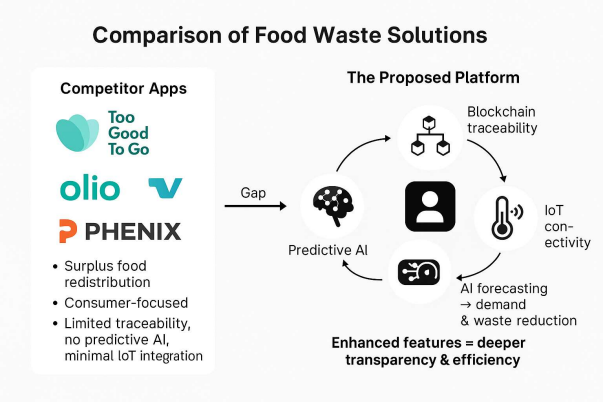
Target Market Analysis

The platform is targeted to many different users, including homes, restaurants, grocery stores, non-profits, and composting organisations. The early adopters are likely to be urban customers who are environmentally conscious and companies whose primary goal is to promote sustainability. This is aimed at bringing these fragmented stakeholders together to encourage them to work together towards better food redistribution, and also in making better decisions which are more informed and are based on data [5].

Competitor Analysis & Technological Gap:

Although it is popular, other competitors such as Too Good To Go, Olio, and Phenix do not have advanced traceability business features with blockchain, predictive AI, and IoT connectivity. Conversely, the proposed technology presents sensor information to ensure that food quality, blockchain safe food tracking, and machine learning accurate prediction are introduced. This also makes the system highly transparent, besides being more technologically advanced [6].



Marketing Mix (4Ps)

Product: A blockchain and AI-powered platform that provides traceability and predictive analytics.

Price: Retailers and restaurants can choose among subscription options, while households and NGOs can adopt a freemium approach.

Place: Easily connected with NGO and retailer systems, accessible on mobile and web platforms.

Promotion: centred on digital marketing campaigns, EU green programs, and sustainability alliances.

Route to Market & Customer Acquisition:

In the first round, a pilot initiative in Berlin, Paris, and Amsterdam will be conducted in collaboration with eateries and non-governmental organisations. Phase two of the strategy calls for expansion through partnerships with local government and EU financing. In order to increase user and beneficiary trust and awareness, we will also make advantage of CSR collaborations and sustainability certifications.

Evidence

Quantitative Data:

- 59 million tons of food waste are generated annually, leading to €132 billion in losses [2].
- The food waste management market is expected to grow at a 5.7% annual rate through 2029[3].
- The app market in this sector is projected to grow at 16.8% annually from 2024 to 2034[2].

Qualitative Insights:

- Digital waste-tracking tools can help save up to €8,300 per kitchen each year (ScienceDirect, 2024).

Data Sources:

Marketdata Forecast, IMARC Group, Europea, and EEA [3][4][5]

Financial Outlook & Viability:

The platform may become profitable by obtaining just 1% of the approximately €264 million EU food donor market through subscriptions and analytics services. EU sustainability grants will assist finance early R&D and reduce entry barriers, while B2B licensing and data analytics will provide the initial revenue.

Conclusion & Recommendations:

Europe's food waste problem offers a financial opportunity as well as an environmental one. The suggested blockchain-secured, AI-powered system has a number of advantages over current options in terms of scalability, intelligence, and transparency. Launching a pilot program, growing alliances, submitting grant applications, and consistently enhancing AI skills are important next steps [6].

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